

NANDINI SHARMA

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Professional Summary

M.Sc. Economics (Data Analytics) student with hands-on experience in econometric modelling, machine learning, and financial data analysis. Built end-to-end projects spanning credit risk prediction (XGBoost, SHAP, ROC-AUC 0.7624), banking sector analysis (OLS, Newey-West HAC), and macroeconomic forecasting (ARIMA, ARIMAX, SQL, Power BI). Seeking roles in Financial Analytics, Economic Research, Data Analytics, or Credit Risk where rigorous quantitative methods drive real-world decisions.

Education

M.Sc. Economics (Data Analytics Specialisation) <i>Symbiosis School of Economics, Pune</i>	2025 – 2027
B.A. (Hons.) Economics <i>JERC University, Jaipur</i>	2022 – 2025 CGPA: 9.07

Technical Skills

Programming & Analytics	Python (Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Statsmodels, Matplotlib, Seaborn), SQL (SQLite, JOINS, CTEs, Window Functions)
Machine Learning	Logistic Regression, Random Forest, XGBoost, SHAP (Model Explainability), Class Imbalance Handling, ROC-AUC, Confusion Matrix
Tools & Platforms	Power BI, Excel (Advanced), Jupyter Notebook, VS Code, GitHub, SQLite

Work Experience

Treasury Intern – SK Finance	<i>8 Jun 2026 – 8 Jul 2026</i> <i>Rajasthan</i>
- Reviewed corporate bond holdings and treasury cash-flow positions; flagged risk-return outliers for the treasury desk - Compiled fixed-income reporting used in weekly investment committee updates	
Business Analyst Intern – Outlook Magazine	<i>June 2024 – Aug 2024</i> <i>Ahmedabad, Gujarat</i>
- Performed market and industry analysis using secondary data to support business reports - Compiled and interpreted datasets underpinning analytical articles and features	
Research Trainee – Budget Analysis and Research Centre	<i>Jun 2023 – Aug 2023</i> <i>Jaipur, Rajasthan</i>
- Conducted detailed research on government budget allocations across sectors to assess spending priorities - Prepared analytical reports and policy briefs supporting budget evaluation and fiscal analysis - Assisted in evaluating subsidy schemes by examining coverage, effectiveness, and allocation efficiency	

Projects

Loan Default Prediction — Home Credit Default Risk	<i>Python, XGBoost, SHAP, Scikit-learn</i>
- Built an end-to-end credit risk model on 307,511 loan applicants achieving ROC-AUC of 0.7624 using XGBoost - Engineered 19 features from application, bureau (1.7M records), and previous application (1.67M records) data - Applied SHAP values to explain model predictions — identified EXT_SOURCE scores as strongest default predictors - Backtested on holdout data: flagged 65% of defaulters at chosen threshold, translating to a simulated Rs. 1.16 billion reduction in loss exposure vs. no-model baseline - Handled class imbalance (8.07% default rate) using <code>scale_pos_weight</code>; compared LR, Random Forest, and XGBoost	
Banking Sector NPA Analysis — PSB vs Private Banks (1997–2024)	<i>Python, OLS, RBI Data</i>
- Analysed Gross NPA ratios for Public and Private Sector Banks over 27 years using RBI official data - Applied OLS regression with Newey-West HAC standard errors to correct for severe autocorrelation ($DW = 0.18$) - Conducted full diagnostic battery: ADF, KPSS, Breusch-Godfrey, Jarque-Bera, ACF/PACF tests - Found macro credit indicators insignificant for PSB NPAs — evidencing institutional and governance drivers	
Fiscal Deficit Forecasting — India (1980–2014)	<i>Python, SQL, ARIMA, ARIMAX, Power BI</i>
- Forecasted India's fiscal deficit-to-GDP ratio using ARIMA(0,1,0) and ARIMAX models on 34-year RBI dataset - Performed SQL data engineering using multi-table JOINS, CTEs, <code>LAG()</code> window functions, and moving averages - Identified and corrected spurious regression through stationarity testing and first-differencing - Holdout validation (2009–2014): RMSE = 0.567, all actual values within 95% confidence interval - Built interactive Power BI dashboard for fiscal trend visualisation and policy communication	

Certifications

- IBM Data Analyst Professional Certificate — IBM / Coursera
- NISM Common Derivatives — National Institute of Securities Markets

Leadership & Extra-Curriculars

- Economics Representative, Upscale Society — organised seminars, events, and departmental symposiums; Content Writer, Upscale (Oct 2023 – Present)